

COMPANY PROFILE 2026 · DIGITIZATION CAPABILITY STATEMENT

Digital systems for government-scale climate and development delivery

4D Climate Solutions designs, builds, and operates secure digital information systems for ministries, development partners, and communities across Southern Africa — from national compensation registries to AI-powered advisory platforms, built offline-first for African realities.

Prepared for ministry submission — in support of the Government of Lesotho's digitization agenda, 2026.

A Lesotho-grown digital and climate solutions firm

4D Climate Solutions (4D) was founded in 2018 by three climate action enthusiasts in Lesotho, driven by a mission to empower communities to engage in solutions to combat climate change and strengthen their resilience. What sets 4D apart is **how** we deliver: cutting-edge digital technology combined with deep community engagement, ensuring sustainable and lasting impact.

Over seven years, digitization has become the backbone of our work. We have moved government and development programs from paper registers, spreadsheets, and manual verification to secure, auditable, real-time digital systems — designed for low-bandwidth environments, local languages, and field conditions. Today 4D operates live production platforms for public-sector programs, including the compensation information system serving the Lowlands Water Development Project.

4D AT A GLANCE

7+ YEARS OF IMPACT	50+ PROJECTS DELIVERED	8 DISTRICTS COVERED	5 COUNTRIES OF EXPERIENCE
8+ DIGITAL APPLICATIONS BUILT	45,000+ PLATFORM USERS (PILOT)	M10+ CONTRACT VALUE	20+ PARTNER ORGANIZATIONS

OUR MISSION

To empower communities to actively engage in solutions to combat climate change and enhance their resilience — using digital tools that put reliable information in the hands of government, field teams, and citizens.

WHY IT MATTERS FOR DIGITIZATION

We do not only write software — we run the community mobilization, training, and change management that make digital systems stick. Every 4D platform ships with capacity building, data governance, and a sustainability plan.

Digitization capabilities

Six capability pillars cover the full lifecycle of a government digitization program — from system design and data migration to field rollout, training, and long-term operation.

-
- 01 Government information systems & registries**
- Design and development of secure management information systems for public programs: beneficiary and asset registries, compensation management, grievance redress, monitoring & evaluation, and administrative workflows. Systems include role-based access control, approval chains, full audit trails, and print-ready official forms.
-
- 02 Field data collection & digitization of paper records**
- Offline-first mobile and web applications for surveys, censuses, verification exercises, and community consultations (pitsos, FGDs, KIIs) — with GPS capture, photo and document digitization, and real-time dashboards for supervisors. Built to work on 2G/3G networks and sync automatically when connectivity returns.
-
- 03 AI & smart data systems**
- AI-powered decision support and advisory platforms through our dedicated 4D AI division (ai.4dcs.co.za): retrieval-augmented (RAG) advisory systems, AI vision for real-time monitoring, and predictive analytics — engineered for low-bandwidth environments, local languages, and African realities, using Anthropic Claude AI.
-
- 04 Secure cloud infrastructure & data protection**
- Cloud-hosted architectures with authenticated access, database row-level security, encrypted document storage, allow-listed user registration, and immutable change logs — so every edit to a public record is attributable and reviewable.
-
- 05 GIS, remote sensing & IoT**
- Geospatial analysis, remote-sensing assessments, and IoT sensor networks — from groundwater aquifer classification (with the University of Virginia) to solar-powered smart irrigation with live soil-moisture telemetry.
-
- 06 Digital capacity building & adoption**
- Tailor-made training for government staff, field teams, and communities; formation of community structures; user onboarding, help materials, and phased handover — ensuring systems are owned and sustained locally after project close.

CIMS — Compensation Information Management System

A full digitization of asset registration, verification, valuation, and compensation tracking for Project Affected Persons (PAPs) under the **Lesotho Lowlands Water Development Project Phase III (LLWDP III)** — replacing paper files and manual reconciliation with a single, secure, auditable source of truth.

Client context: LLWDP III (African Development Bank–financed)

Status: Live — cims.4dcs.co.za

Developed & operated by: 4D Climate Solutions

1,388

PAP RECORDS
DIGITIZED

94%

RECORDS FIELD-
VERIFIED

25

ROUTES (17 RURAL · 8
URBAN)

16

COMMUNAL ASSET
CATEGORIES

WHAT CIMS DIGITIZES

- **Household & asset registry.** Beneficiaries, co-owners, land parcels, structures, and communal assets in one linked database.
- **Verification workflow.** Print-ready two-page verification forms with photos and ID documents; field completion writes straight back to the database.
- **Compensation valuation.** A central Rates Master governs land and asset rates; compensation auto-calculates (area × rate + disturbance) — one source of truth, no spreadsheet drift.
- **Payment tracking.** Payment status and payment documents per PAP, with live dashboard counters for management oversight.
- **Document management.** IDs, photos, and payment evidence uploaded from the field to encrypted cloud storage, linked to each record.

GOVERNANCE & DATA INTEGRITY

- **Role-based access.** Distinct tiers for administrators, community liaison and resettlement officers, field surveyors, and view-only client users.
- **Approval chain.** Ordinary edits flow through an edit-request and approval workflow rather than direct changes to official records.
- **Live audit trail.** Every change is logged and visible in a real-time Change History on each record.
- **Row-level security.** Database-enforced permissions on every table; registration restricted to an authorized email allow-list.
- **De-duplication.** Guided merge workflows resolve duplicate PAP files — including multi-asset households — with a reviewable result before commit.

TECHNOLOGY

React single-page application · PostgreSQL (Supabase) with row-level security and realtime subscriptions · Express API · Cloudflare R2 encrypted object storage · continuous deployment on Render. Delivered with user training, operational handover notes, and live production support.

Platforms built and operated by 4D

Eight-plus custom applications delivered for government programs, development partners, and local enterprises — designed offline-first for low-bandwidth environments.

Lipalo — AI Agricultural Advisory

LIVE

Lesotho's AI-powered agricultural advisory system (v6.8.0), bridging a 1:1,500 extension-officer-to-farmer gap. Built on Anthropic Claude AI with 101 agronomy knowledge chunks and 40+ API endpoints.

- Real-time weather alerts with farming impact analysis
- FAO-based crop calendars for 4 agro-ecological zones
- LESIS-based soil analysis with crop suitability ratings
- FAO-56 irrigation scheduling — 8 crops, all 10 districts

LLWDP II Monitoring App

LIVE

Field data collection for community mobilization under the Lowlands Water Development Project Phase II — digitizing consultations end-to-end.

- Offline-first PWA architecture for remote districts
- Pitsos, FGDs, KIIs, and baseline surveys
- GPS location capture on every activity
- Real-time administrative dashboard

Letlotlo (Gooi Gooi Savings)

LIVE

Financial inclusion platform digitizing savings groups, burial societies, street vendors, and factory workers — with Business Apps, KTB, and MOB.

- 45,000+ users across 2,600+ groups (pilot)
- VSLA, burial society & group management
- Mobile money integration & loan access facilitation

Rekisa POS

LIVE

Multi-tenant point-of-sale system digitizing sales and inventory for Lesotho businesses.

- Inventory management, sales tracking & reports
- Multi-branch support
- Offline transaction sync

Maseru Bridge Traffic Bot

IN DEVELOPMENT

AI-powered traffic monitoring for the Maseru Bridge border crossing.

- Real-time queue status via AI vision (~500 image analyses daily)
- WhatsApp integration for travelers
- Historical traffic patterns

GrowSmart IoT Irrigation

IN DEVELOPMENT

Solar-powered, IoT-based smart irrigation implemented with LECSA in Morija.

- Live soil-moisture telemetry & weather integration
- Remote pump control from a mobile app
- Predictive analytics & personalized crop advice

Kopanya Fresh

IN DEVELOPMENT

Agricultural marketplace connecting smallholder farmers with urban buyers.

- Three-tier crop pipeline tracking
- Buyer, Farmer & Agent portals
- Order management & delivery coordination

Rekisa Marketplace

IN DEVELOPMENT

AI-powered fresh produce marketplace with distinct user roles.

- Buyer, Seller, Field Supervisor & Admin roles
- Real-time produce listings, quality grading & pricing
- Delivery coordination

Delivering for government & development partners

Selected current and completed assignments — with the digitization component of each engagement highlighted.

PROJECT	CLIENT / PARTNER	DIGITIZATION ELEMENT
LLWDP III — Asset Verification & Compensation GOV Oct 2025 – Feb 2026	African Development Bank—financed; community mobilization, RAP implementation & grievance redress in Botha-Bothe & Leribe	CIMS — live digital registry of 1,388 PAPs with verification, valuation & payment tracking (see page 4)
LLWDP II Community Mobilization Study GOV Oct 2025 – Feb 2026	Ministry of Natural Resources · JV with Khanyisa Projects	Digital monitoring system for all field data collection — pitsos, FGDs & surveys captured on an offline-first app with live dashboards
SADP II National Service Provider — Southern Region GOV May 2025 – Jun 2026	Ministry of Agriculture, Food Security & Nutrition · World Bank (IDA 64220)	Digital administration of matching-grant beneficiaries and field service provider management
Lipalo — AI Agricultural Extension Ongoing	University of Virginia · National University of Lesotho	National-scale AI advisory platform — weather, crop calendars, soils & irrigation across all 10 districts
LHWP Climate Change Vulnerability Assessment Aug 2022 – Dec 2023	Lesotho Highlands Development Authority	Data-driven vulnerability assessment and adaptation/mitigation strategy for LHWP Phases I & II
Agricultural Insurance Feasibility Study Feb – Dec 2020	IFC (World Bank Group) · RMSI India	Market and data analysis for smallholder agricultural insurance across six value chains
Value of Water Study Ongoing	University of Virginia	Economic valuation of water resources in highland communities, incl. community-led digital soil & water monitoring
Gooi Gooi Savings Solutions Ongoing	Business Apps · KTB · MOB	Digitized savings system promoting financial inclusion for street vendors and factory workers
GrowSmart Solar Irrigation Ongoing	LECSA (Moriya)	IoT sensor network, predictive analytics & mobile-app control for climate-smart irrigation

Our approach to public-sector digitization

Principles proven across our live government systems — the same standards we would bring to the Ministry's digitization project.

BUILT FOR THE FIELD, NOT JUST THE OFFICE

Offline-first applications that keep working on 2G/3G networks and sync when connectivity returns; GPS, photo, and document capture at the point of service; print-ready official forms where paper is still required.

SECURITY & ACCOUNTABILITY BY DESIGN

Role-based access, database row-level security, allow-listed registration, approval workflows for changes to official records, and immutable audit logs — every edit attributable, every record reviewable.

ONE SOURCE OF TRUTH

Central master data (rates, categories, registries) drives every calculation and report, eliminating spreadsheet drift and reconciliation disputes between departments, contractors, and communities.

AI WHERE IT MULTIPLIES CAPACITY

Applied AI through our 4D AI division — advisory chat, AI vision monitoring, and predictive analytics — designed for local languages, low bandwidth, and responsible public-sector use.

ADOPTION, NOT JUST DEPLOYMENT

Community sensitization, staff training, user onboarding, and phased handover with documentation — the change management that determines whether digitization sticks.

COST-EFFECTIVE & LOCALLY SUSTAINED

Lean cloud architectures with modest running costs; a flexible staffing model that mobilizes specialist expertise per engagement; systems documented for long-term local operation.

TECHNOLOGY & DELIVERY PARTNERS

TECHNOLOGY

- Anthropic — Claude AI engine for Lipalo & advisory systems
- Supabase — cloud database & authentication infrastructure
- Cloudflare — encrypted object storage & delivery
- Render — cloud hosting & continuous deployment
- SunMoksha / G8Way Technologies — IoT & hardware

DELIVERY & RESEARCH

- Khanyisa Projects — joint venture partner
- University of Virginia (USA) · Bangor University (UK)
- National University of Lesotho (NUL)
- RMSI India Pvt. Ltd.
- Business Apps (Pty) Ltd. / OOWEBOO Web Engineers

Leadership & team

A lean core leadership team, supported by a dynamic network of 20+ specialized project staff mobilized per engagement — from data scientists and GIS analysts to community liaison officers.

Ehsan Rizvi

DIRECTOR

A development professional with over 20 years in senior management roles across climate change, food security, natural resource and watershed management, agricultural livelihoods, social protection, and cash-based emergency response — with programs in India, Afghanistan, Sri Lanka, Tanzania, and Lesotho. Ehsan leads the design and implementation of 4D's digital systems, has mobilized significant resources for land restoration, implemented community-led soil and water monitoring systems, and collaborated with the University of Virginia on groundwater aquifer classification. He holds a PG Diploma in Food Security & Climate Change from Bangor University, UK, and a B.Tech in Civil Engineering from IIT Kanpur.

Dr. Poloko Mosebi

CLIMATE CHANGE EXPERT

PhD in Plant Science and MSc in Pasture Science (University of Pretoria). Leads key initiatives including LLWDSP III resettlement efforts and the GrowSmart initiative, integrating advanced technology with sustainable agricultural practices.

Bartholomew Mofolisa

PROGRAM COORDINATOR

Over 20 years of experience in livelihood projects, market system development, and human-centered supply chains across Lesotho.

TECHNICAL STAFF

- Mantsatsi Moleleki — Soil and Water Expert
- Fusi Mosebo — M&E Officer
- Makhotso Mohoanyane — Social Expert
- Momakuenta Matebesi — Resettlement Expert
- Palesa Mabaso — Administrative Assistant

EXTENDED TEAM EXPERTISE

- Data Science & AI Engineering · ICT for Development (ICT4D)
- Remote Sensing & GIS Analysis
- Environmental & Social Assessments
- Gender & Social Inclusion · Community Liaison
- Financial Inclusion · WASH & Health · Agricultural Extension

INSTITUTIONAL CLIENTS & PARTNERS

GOVERNMENT & PUBLIC SECTOR

- Ministry of Natural Resources, Lesotho
- Ministry of Agriculture, Food Security & Nutrition, Lesotho
- Lesotho Highlands Development Authority (LHDA)
- Government of Lesotho

LOCAL & REGIONAL

- Caritas Lesotho · Care for Basotho Association
- Khathang Tema Batšukuli (KTB) / Mme-O-Bohlokoa (MOB)
- LECSA — Lesotho Evangelical Church in Southern Africa

INTERNATIONAL DEVELOPMENT

- World Bank / International Finance Corporation (IFC)
- African Development Bank (AfDB)
- Millennium Challenge Corporation (MCC)

COMPANY REGISTRATION

- Incorporated May 2018 · single-shareholding company
- License no. TR55456 — Ministry of Trade and Industry, Cooperative and Marketing
- Registered under the Industrial License Act 1969 & Trading Enterprise Act 1993

LET'S BUILD IT TOGETHER

Ready to partner with the Ministry on digitization that works in the field — and lasts.

From CIMS to Lipalo, 4D has shown that world-class digital public infrastructure can be designed, built, and operated from Maseru. We would welcome the opportunity to demonstrate our live systems and discuss how this experience can serve the Ministry's digitization project.

VISIT US

Office No. B15, 3rd Floor
Maseru Roller Mills Building
Moshoeshoe Road, Industrial Area
Maseru 100, Lesotho

REACH US

Email: info@4dcs.co.za
Phone: +266 2232 3442 · +27 76 108 0024
WhatsApp: +27 76 108 0024
www.4dcs.co.za · AI Division: ai.4dcs.co.za